

3 PREPARATION OF BENCH MATERIAL FOR EXCAVATION

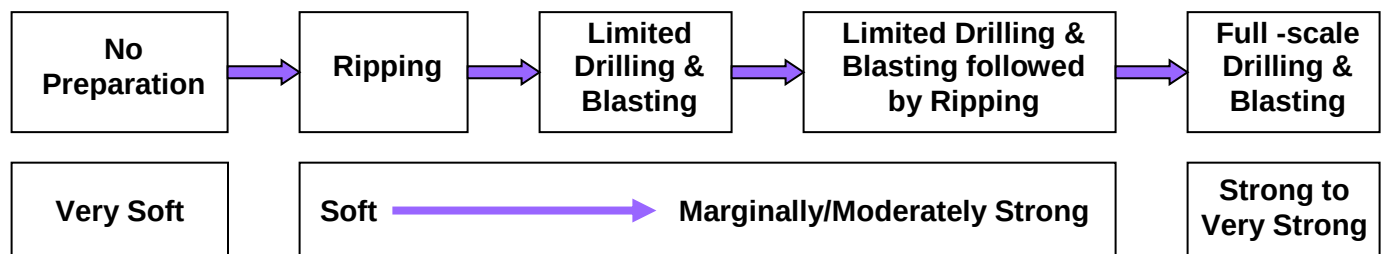
Preparation of bench material involves the processes of changing the mechanical state of in-situ rock-mass of the bench, wherever necessary, so that it becomes amenable to easy extraction by the excavators.

The purpose of preparing the rock-mass for extraction is to ensure safety in mining, the production of raw material of the required quality, and the possibility of using means of mechanization in the best possible operating conditions.

The methods used for bench material preparation are mainly by Ripping (mechanical loosening by actuating members) and Drilling & Blasting. However, hydraulic (injection, water saturation, dissolution, etc.), physical (electromagnetic and thermal action), chemical and combined methods may also be successfully employed under certain conditions.

The method of bench material preparation is chosen based on the in-situ state of the rock-mass (state of aggregation, strength, presence and orientation of weakness planes, etc), capacity of the mining plant, the available technical means, and the mining system adopted.

When the material is soft enough, it does not require any preparation as it can be directly excavated and loaded onto transport equipment by excavators. If the material is marginally strong, limited drilling and blasting (without any throw) may be used to loosen the ground. For moderately strong rocks, limited drilling and blasting followed by ripping is used sometimes as means of material preparation. Full scale drilling and blasting is used in rock-mass that cannot be excavated directly or cannot be loosened effectively/ economically by ripping (with or without limited drilling and blasting).



3.1. Ripping

Ripping is a method used for preparation of bench material / rock-mass that are soft to marginally hard to facilitate subsequent loading and transport through mechanical loosening. Ripping with dozers as a means of loosening the rock-mass came into existence in late 1950s and has become a popular method. Ripping is usually economic than drilling and blasting as a means of bench material preparation. But as the rockmass becomes harder drilling and blasting becomes cheaper.

Ripping is done by ripper that is basically an attachment having single tooth or multiple teeth (up to five teeth) fitted to a shank. Ripper is generally attached at the rear of a dozer (tractor-mounted) and operated hydraulically. Weathered igneous and metamorphic rocks are generally prepared with the use of single tooth-rippers. For loosening the dense rocks generally multi-teeth rippers are used to raise their output. The depth of ground penetration of tractor-mounted rippers is varied by means of hydraulic control system. Low or medium jointed weathered igneous and

metamorphic rock-mass are ripped with tooth/teeth having straight uprights. Heavily jointed and/or brittle rock-mass are ripped with tooth/teeth of intricate shape.

With the evolution of high power ripper-dozers, the rock-mass that were considered earlier to be non-rippable now comes under rippable category. As it stands today ripping, while taking care of environment and safety, have become competitor to drilling and blasting for bench material preparation.

3.1.1. Classification, Applicability and Limitations

Classification of Rippers

Rippers may be classified into mainly two types – tractor-mounted (as an attachment to dozers) rippers and tractor-trailed rippers. The tractor mounted rippers are hydraulically operated and positive force can be applied for tooth/teeth penetration. But in case of tractor-trailed rippers the tooth/teeth penetration becomes a function of the shank weight, the sharpness of the tooth/teeth and the hardness/strength of the rock-mass being ripped. However, in surface mining applications the use of tractor-trailer type rippers has become more or less obsolete.

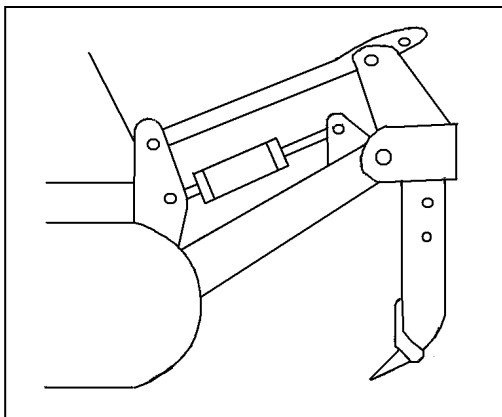
Rippers are also classified into two groups based on the number of tooth – single-tooth rippers and multi-teeth rippers. The hydraulically operated tractor mounted rippers are generally available in two designs – parallelogram design and adjustable design.



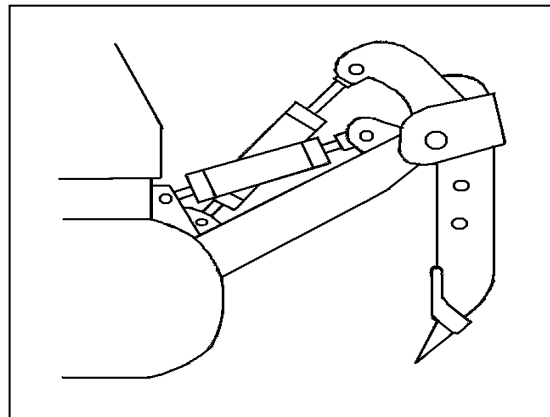
SINGLE TOOTH RIPPER



MULTI TEETH RIPPER



Parallelogram design



Adjustable design

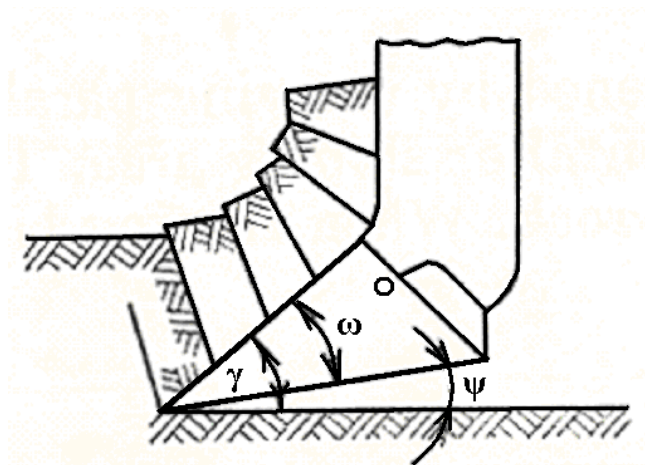
DESIGNS FOR HYDRAULICALLY OPERATED RIPPERS

Applicability and Limitations of Rippers

Rippers are generally used in soft to marginally hard rock-mass. It is mostly used to excavate overburden, but can also be used for soft to marginally hard ore body. If the rock-mass is bit more hard, some times it is first weakened by limited drilling and blasting followed by ripping. The rippers do not find much application in hard or very hard rocks.

3.1.2. Working Principle of Rippers

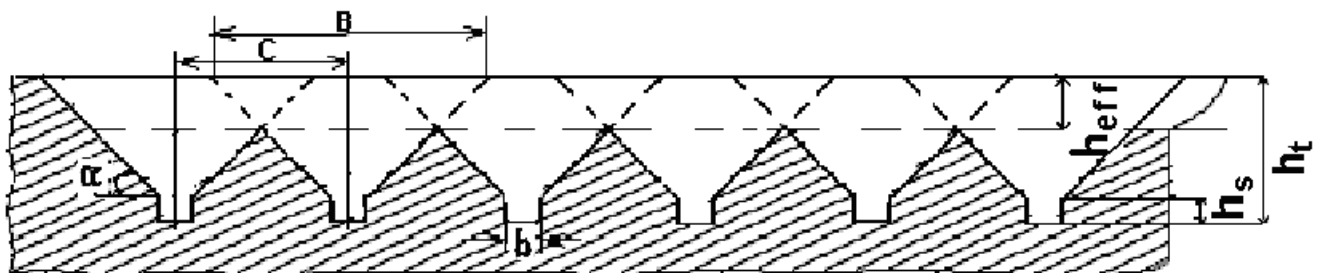
The geometry of the working member of a ripper includes – cutting angle γ , tip-edge angle ω , clearance angle ψ , tooth thickness and length, and tooth spacing (in case of multi-teeth ripper).



WORKING TOOL OF A RIPPER

The cutting force of a ripper depends on the ripping angle. For weathered igneous, metamorphic and frozen rock-mass the optimum ripping angle is about 30 to 45 degrees. An increase in the ripping angle from 40 to 60 degrees approximately doubles the resistance to ripping. A decrease in the ripping angle to less than 30 degrees also increases the resistance to ripping.

The tip-edge angle ω generally varies between 20 to 30 degrees. Its magnitude should be such that the clearance angle $\psi \geq 5$ to 7 degrees when loosening igneous and metamorphic rock-mass (compact or weathered), and $\psi \geq 8$ to 10 degrees while dealing with frozen rock-mass. A smaller clearance angle causes crushing of the rock by the back face of the tip, and this increases the tooth wear and resistance to ripping.



RIPPER CUTS IN PARALLEL PASSES

When a ripper starts ripping, the rock-mass breaks up within the boundaries of a trapezium shaped cut. When rock-mass are loosened a slot forms in the bottom part of the cut with a base width (b) that is close to the thickness of the tooth tip (b_1) and a slot height (h_s) that is equal to 0.15 to 0.2 times the ripper tooth penetration (h_t) into the ripped rock-mass. The slope of the side walls of the cut (α) varies from 40 to 60 degrees depending on the rock-mass characteristics and the tooth tip geometry.

The rock-mass loosening ability of the ripper is specified by the possible penetration (h_t) of the ripper tooth into the loosened rock-mass that depends on rock-mass properties and the jointing pattern of the rock-mass. Monolithic rock-mass undergo loosening due to overcoming of their tensile strength, whereas in case of jointed rock-mass it is due to overcoming the cohesion between the structural blocks. The efficiency of ripping increases when it is applied to rock-mass with a natural jointing or developed bedding.

3.1.3. Method and Cycle of Operation, Estimation of Productivity

Method and Cycle of Operation of Ripper

A ripper loosens rockmass through carrying parallel adjoining passes on a horizontal or sloping area. This results in the creation of a layer of broken rock. In ripping horizontal surfaces/benches, working passes of the ripper are performed in a shuttle-type way. When ripping is carried out in sloping (up to 20°) surfaces, ripping is done in down grade direction only and it travels idle in upgrade.

The distance between adjacent passes (C) is set aiming at ensuring the required lump size and a sufficient loosening depth. When the ripper carries out parallel passes, zones of non-ripped rock are formed between adjacent cuts in the bottom part of the cross section that hamper the extraction of the loosened material upto the full depth of tooth penetration. The depth of effective ripping (h_{eff}) is less than the full depth of tooth penetration (h_t) into the ground, and h_{eff} depends on tooth geometry, effective angle of breaking (α) that depends on the rock-mass properties, distance between adjacent passes (C) and full depth of tooth penetration (h_t). To increase h_{eff} , supplementary crossing passes of the ripper at right angles or diagonal to the first passes are carried out to break the pillars and to ensure lumps of rock of a better size. The spacing between the crossing passes (C') is generally kept at 1.2 to 1.5 times the C .

The loosening of rock-mass depends on the direction of ripping with respect to the system of joints. Ripping at right angles to the direction of the main jointing proves to be most effective. When laminated weathered igneous and metamorphic rock-mass are ripped, the initial penetration of the tooth into the ground is of greatest importance. To facilitate the penetration, an "advanced cut" of required depth is made at right angles to the contemplated parallel passes of the ripper by carrying multiple passes of the ripper or by blasting.

The parameters of power ripping are determined by the amount of possible tooth penetration h_t .

The top width of a single furrow is given by the following formula.

$$B = 2(h_t - h_s) \cot \alpha + b$$

The quantity B increases 1.5 to 2.0 times when the ripper is fitted with a cutter wing secured to the upright.

LECTURE NOTES – SURFACE MINING

In parallel passes of the ripper, the rock-mass is effectively loosened to a depth specified by the following formula.

$$h_{\text{eff}} = (h_t - h_s) - ((C - b) \tan \alpha / 2)$$

The value of the base width (b) depends on the value of the thickness of the tooth tip (b_1) and the joint condition of rock-mass as given below.

Values of Width of Cut at the Base (b)

Joint condition of rock-mass	Acoustic index of rock jointing ($A_j = v_m^2 / v_l^2$)	b
Low jointing	0.60 to 0.90	(1.5 to 2.0) $\times b_1$
Medium jointing	0.40 to 0.60	(2.0 to 3.5) $\times b_1$
High jointing	< 0.40	(3.5 to 6.0) $\times b_1$
❖ v_m and v_l are the velocities of longitudinal elastic waves in rock-mass and in rock lump respectively		
❖ b_1 is the tip width of the ripper tooth		

When the ripper makes crossing passes with the spacing between its adjacent passes (C') equal to 1.2 to 1.5 times the spacing between first parallel passes (C), the depth of effective loosening becomes almost equal to tooth penetration depth i.e. $h_{\text{eff}} \approx h_t$.

Estimation of Productivity of Ripper

The output of a ripper may be estimated from the following relations.

$$Q_r = \frac{3600 \times C \times h_{\text{eff}} \times K_u}{\frac{1}{v_r} + \frac{T}{L}} \quad \text{m}^3/\text{hour, in parallel passes} \quad [1]$$

$$Q'_r = \frac{3600 \times h_t \times K_u}{\frac{1}{v_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + T \left(\frac{1}{C \cdot L} + \frac{1}{C' \cdot L'} \right)} \quad \text{m}^3/\text{hour, in parallel and crossing passes}$$

where, K_u (= 0.7 to 0.8) is the ripper utilization factor; v_r is the technical ripping velocity in m/sec, T is the time of ripper travel to the next furrow in sec, and L and L' are the length of parallel and crossing pass respectively in m.

The output of rippers used on dense rocks may reach 1000 to 1500 m^3/hour . However, It depends on the length of parallel cuts and it is convenient to set the length within 100 to 300 m range.

Mechanical, or power, loosening facilitates selective extraction of thin horizontal and inclined (up to 20°) beds, permits effective control of lump size, makes it possible to reduce the waste and dilution of the mineral and intermixing of rocks. As a means of bench material preparation, it is safer and environmentally more acceptable compared to drilling and blasting.

Rippers are successfully used in mining bauxite, coal, phosphate and apatite ores, shales, sandstones, weathered limestone and also thin slices of highly and extremely jointed ores and rocks. Power ripping is effective when heavy clay rocks and frozen rocks are worked by means of hydraulic and also for auxiliary operations (making drainage ditches, stubbing, etc.). Proper preparation and the presence of a loosened slice of small thickness permit extraction of the mined rock with the aid of scrapers, bulldozers and single-bucket loaders.

3.1.4. Concept of Rippability

Rippability of a rock-mass is the measure of ease or difficulty with which it may be ripped by a ripper. The rippability of a rock-mass is a function of the rock-mass characteristics including the joint (or other weakness plane) patterns and the ripper power.

The best way of rippability assessment and of equipment investment analysis is through actual field trial at the site of ripping. However, trials are not always practicable and therefore the need arises for indirect methods, all of which involve correlation with various rock-mass/rock sample parameters. Even in cases where trial demonstration is possible, indirect methods would still be necessary because the geo-technical properties of the rock-mass change quite often within a short distance.

Earlier ripper-dozer manufacturers were recommending the use of seismic wave velocity (P wave) as the single rock parameter for evaluation of feasibility of ripping a rock mass and also to predict the ripper performance, capital and operating costs. With the introduction of high powered rippers it was realised that seismic wave velocity alone was not sufficient as other geotechnical parameters particularly the joint spacing are of equally dominating influence.

Weaver (1975) was the first to propose the 'rippability rating' system based on summing weighted rock-mass parameters similar to the rock-mass rating (RMR) system of South Africa. He introduced 'seismic velocity' as the main parameter by replacing two parameters of RMR namely the intact rock strength and the RQD.

Smith (1986) modified the Weaver rippability rating (RR) parameters and omitted the seismic velocity from the rating. The RR proposed by Smith does not include the seismic velocity as a directly considered parameter and he proposed that seismic velocity of rock-mass can be independently used for the purpose of comparison with RR. He proposed a correlation of the RR with seismic velocities under average conditions with expected ripping production or given tractor selection.

It is appropriate here to mention that out of six parameters considered by Smith for estimating RR, at least five parameters excepting the rock hardness are difficult to estimate at the pre-excavation stage. This is one of the limitations of RR. To overcome the Smith's drawback, Singh et al (1986, 1987) proposed an alternative Rippability Index Classification (RIC) and tested it in a number of case histories in England and Turkey.

Singh et al (1987) reviewed some of the well known ripper attachable dozers available and proposed a correlation between RIC, dozer power (kW) and weight (tonne) based on performance of rippers in British surface coal mines. According to them the rock-mass is expressed as rippable when tractor ripper with a single shank attachment penetrates more than 0.60 m and obtains a minimum forward speed of approx. 2.5 km/hour. The main findings of Singh et al are - (i) rock abrasiveness controls both ripper life and production rate, (ii) some Classification parameters such as discontinuity spacing can only be determined if rock units are exposed and (iii) further studies are required to assess rippability where rock units are not exposed.

LECTURE NOTES – SURFACE MINING

The geotechnical parameters that influence the rippability of rock-mass mainly include the following.

- (i) Geotechnical features of the rock mass that include:
 - (a) Nature of formation including presence and spacing of clear planes of stratification/discontinuity/joints and/or absence of substantial fractures or weak planes. Sedimentary rocks are more amenable to ripping as compared to igneous and metamorphic rocks
 - (b) Presence of faults, fracture, joints and planes of weakness at any kind
 - (c) Degree of weathering
 - (d) Brittleness and crystalline nature of rock
 - (e) Degree of stratification and lamination of rock
 - (f) Grain size in the rock-coarse the grain size more amenable the rock for ripping
 - (g) Joint spacing, joint continuity, joint gauge material and strike/dip orientation of joints

It is appropriate to mention here that estimation of these geotechnical features to a fair degree is difficult at the pre-mining stage.

- (ii) Physico-mechanical properties of rocks that include:
 - (a) Compressive, shear and tensile strengths of rock samples
 - (b) Abrasiveness
 - (c) Influence of water on rock strength
 - (d) Moisture content
 - (e) Brittleness and hardness

- (iii) In-situ seismic wave velocity of rock mass

Apart from the above mentioned geotechnical parameters, the rippability of rock-mass also depends on the ripper power. Keeping all other parameters same, if the ripper power is increased the rock-mass becomes more easily rippable.

3.2. Estimation of Number of Blast-hole Drills required in a Mine

The method of estimation of number of blast-hole drills required in a surface mine mining bedded deposits with sharp contact differs from that in case of a surface mine mining massive / vein / pipe type deposits where the ore-body grade change is gradual.

3.2.1. Estimation of Number of Drills required in a Surface Coal Mine

In a surface coal mine (dip of coal seam ≤ 1 in 8) the coal and the overburden and/or inter-burden are excavated in separate benches. Hence, the number of blast-hole drills required for coal bench / benches and that for overburden and/or inter-burden bench / benches are required to be estimated separately.

Some of the following data will be given and the rest are to be assumed calculation for the number of drills required for the mine.

Given Conditions / Assumptions

• Required production of coal – T te / year	• Stripping ratio – SR m ³ / te
• In-situ density of coal – C_d te / m ³	or
• Coal seam thickness – TH_c m	• Overburden thickness – TH_{ob} m
• Coal bench height – BH_c m	• O/B bench height – BH_{ob} m
• Blast-hole pattern for coal bench – $B_c \times S_c$ m ²	• Blast-hole pattern for O/B bench – $B_{ob} \times S_{ob}$ m ²
• Drilling hours / year – H_r	• Sub-grade drilling for O/B bench – SG %
• Drilling rate in coal bench – d_c min / m	• Drilling rate in O/B bench – d_{ob} min / m
• Length of feed in coal bench – f_c m	• Length of feed in O/B bench – f_{ob} m
• Time to move the drill in coal bench – l_c min	• Time to move the drill in O/B bench – l_{ob} min
• Time to set the drill in coal bench – g_c min	• Time to set the drill in O/B bench – g_{ob} min
• Time to change a drill rod in coal bench – e_c min	• Time to change a drill rod in O/B bench – e_{ob} min
• Overall utilization of drill in coal bench – E_c	• Overall utilization of drill in O/B bench – E_{ob}

Estimation of Number of Drills required for Coal Bench/Benches

Height of the coal bench is selected from the given coal seam thickness and the shovel capacity intended to be used. The coal bench height (BH_c) should be so selected that TH_c is an integer multiple of BH_c .

$$\text{Volume of coal to be produced / year} = (T / C_d) \text{ m}^3$$

$$\text{Volume handled / hole} = B_c \times S_c \times BH_c \text{ m}^3$$

$$\text{Total number of holes to be drilled / year} = (T / C_d) / (B_c \times S_c \times BH_c) = N_{hc} \text{ (say)}$$

$$\text{Depth of blast-hole} = BH_c \text{ m, as there is no sub-grade drilling for coal bench}$$

$$\text{Number of feeds required for a hole} = BH_c / f_c = N_{fc} \text{ (say)}$$

LECTURE NOTES – SURFACE MINING

Actual number of feeds required for a hole = N_{fc} , N_{fc} is integer
= (Integer part of N_{fc}) + 1, otherwise

Number of feed change required (N_{cc} say) = $N_{fc} - 1$, N_{fc} is integer
= Integer part of N_{fc} , otherwise

Total time to drill one hole = $((l_c + g_c + (BH_c \times d_c) + (N_{cc} \times e_c)) \text{ min} = t_c \text{ min (say)}$

Actual number of holes drilled per hour per drill = $(60 / t_c) \times E_c$

Actual number of holes drilled per year per drill = $(60 / t_c) \times E_c \times H_r$

Total number of drills required = $N_{hc} / ((60 / t_c) \times E_c \times H_r) = N_{dc}$ (say)

Actual number of drills required = N_{dc} , N_{dc} is integer
= (Integer part of N_{dc}) + 1, otherwise

Estimation of Number of Drills required for O/B Bench/Benches

The stripping ratio is given by –

$$\text{Stripping Ratio (SR)} = \frac{\text{Overburden Thickness (TH}_{ob})}{\text{Coal seam Thickness (TH}_c) \times \text{Density of Coal (C}_d)}$$

If the overburden thickness is given, then the stripping ratio is calculated from the above relation.

$$\Rightarrow SR = TH_{ob} / (TH_c \times C_d)$$

If the stripping ratio is given, then the overburden thickness is calculated as –

$$\text{O/B thickness} = \text{Stripping ratio} \times \text{Coal seam thickness} \times \text{Density of coal}$$

$$\Rightarrow TH_{ob} = SR \times TH_c \times C_d$$

Height of the O/B bench is selected from the given O/B thickness (when it is given) or from the calculated O/B thickness (when stripping ratio is given) and the shovel capacity intended to be used. The O/B bench height (BH_{ob}) should be so selected that TH_{ob} is an integer multiple of BH_{ob} .

$$\text{Volume of O/B to be excavated / year} = (T \times SR) \text{ m}^3$$

$$\text{Volume handled / hole} = B_{ob} \times S_{ob} \times BH_{ob} \text{ m}^3$$

$$\text{Total number of holes to be drilled / year} = (T \times SR) / (B_{ob} \times S_{ob} \times BH_{ob}) = N_{hob} \text{ (say)}$$

$$\text{Depth of blast-hole, } DH_{ob} = BH_{ob} \times (1 + SG/100) \text{ m for sub-grade drilling of SG\%}$$

$$\text{Number of feeds required for a hole} = DH_{ob} / f_{ob} = N_{fob} \text{ (say)}$$

Actual number of feeds required for a hole = N_{fob} , N_{fob} is integer
= (Integer part of N_{fob}) + 1, otherwise

Number of feed change required (N_{cob} say) = $N_{fob} - 1$, N_{fob} is integer
= Integer part of N_{fob} , otherwise

$$\text{Total time to drill one hole} = (l_{ob} + g_{ob} + (DH_{ob} \times d_{ob}) + (N_{cob} \times e_{ob})) \text{ min} = t_{ob} \text{ min (say)}$$

$$\text{Actual number of holes drilled per hour per drill} = (60 / t_{ob}) \times E_{ob}$$

$$\text{Actual number of holes drilled per year per drill} = (60 / t_{ob}) \times E_{ob} \times H_r$$

LECTURE NOTES – SURFACE MINING

Total number of drills required = $N_{hob} / ((60 / t_{ob}) \times E_{ob} \times H_r) = N_{dob}$ (say)

Actual number of drills required = N_{dob} , N_{dob} is integer

= (Integer part of N_{dob}) + 1, otherwise

Numerical Example

Given Conditions:	
Required production of coal – 5 million te / year	Coal seam thickness – 10 m
Stripping ratio – 3 m ³ / te	Dip of coal seam – 1 in 20
Assumptions made:	
Average in-situ density of coal – 1.6 te / m ³	Drilling hours per year – 300 × 2 × 5 = 3000
Drill dia for coal bench – 150 mm	Drill dia for O/B bench – 250 mm
Bench height for coal – 10 m	Bench height for coal – 12 m
Blast-hole pattern for coal bench – 4.5 × 6.5 m ²	Blast-hole pattern for O/B bench – 6 × 8 m ²
	Sub-grade drilling for O/B bench – 10%
Drilling rate in coal bench – 2 min / m	Drilling rate in O/B bench – 4 min / m
Time to move and set the coal drill – 6 min	Time to move and set the O/B drill – 7 min
Time to change one rod in coal drill – 3 min	Time to change one rod in O/B drill – 3 min
Length of feed for coal drill – 7 m	Length of feed for O/B drill – 10 m
Overall utilization of coal drill – 0.6	Overall utilization of coal drill – 0.6
Calculations:	
Total O/B thickness = 3 × 10 × 1.6 m = 48 m ⇒ 4 benches of 12 m height is assumed	
Estimation of number of coal drills required:	
Volume of coal to be handled / year = $(5 \times 10^6) / 1.6 \text{ m}^3 = 3125000 \text{ m}^3$	
Volume handled / hole = $4.5 \times 6.5 \times 10 \text{ m}^3 = 292.5 \text{ m}^3$	
Number of holes required / year = $3125000 / 292.5 = 10683.76 = 10684$	
Depth of hole = 10 m	
Number of feeds required = $10 / 7 = 1.428$	
Actual number of feeds required = $\text{Int}(1.428) + 1 = 1 + 1 = 2$	
Number of feed change required = $2 - 1 = 1$	
Total time to drill one hole = $(6 + (10 \times 2) + (1 \times 3)) \text{ min} = 29 \text{ min}$	
Actual number of holes drilled per hour per drill = $(60 / 29) \times 0.6 = 1.2413\dots$	
Actual number of holes drilled per year per drill = $(60 / 29) \times 0.6 \times 3000 = 3724.13$	
Total number of drills required = $10684 / 3724.13 = 2.86$	
Actual number of drills required = $\text{Int}(2.86) + 1 = 2 + 1 = 3$	

Estimation of number of O/B drills required:
Volume of O/B to be handled / year = $(5 \times 10^6) \times 3 \text{ m}^3 = 15 \times 10^6 \text{ m}^3$
Volume handled / hole = $6 \times 8 \times 12 \text{ m}^3 = 576 \text{ m}^3$
Number of holes required / year = $(15 \times 10^6) / 576 = 26041.66 = 26042$
Depth of hole = $12 (1 + 0.1) \text{ m} = 13.2 \text{ m}$
Number of feeds required = $13.2 / 10 = 1.32$
Actual number of feeds required = $\text{Int} (1.32) + 1 = 1 + 1 = 2$
Number of feed change required = $2 - 1 = 1$
Total time to drill one hole = $(7 + (13.2 \times 4) + (1 \times 3)) \text{ min} = 62.8 \text{ min}$
Actual number of holes drilled per hour per drill = $(60 / 62.8) \times 0.6 = 0.5732...$
Actual number of holes drilled per year per drill = $(60 / 62.8) \times 0.6 \times 3000 = 1719.74$
Total number of drills required = $26042 / 1719.74 = 15.14$
Actual number of drills required = $\text{Int} (15.14) + 1 = 15 + 1 = 16$

3.2.2. Estimation of Number of Drills required in a Surface Mine for extracting massive/ vein/ pipe type deposits

In mines extracting massive/vein/pipe type deposits where the ore waste contact is not sharp but gradual, both the ore and waste are mined in same benches. So the number of drills required is estimated considering the ore and waste together.

Given Conditions / Assumptions

• Required production of ore – $T \text{ te / year}$	• Stripping ratio – $SR \text{ te / te}$
• In-situ density of ore – $O_d \text{ te / m}^3$	• In-situ density of waste – $W_d \text{ te / m}^3$
• Bench height – $BH_t \text{ m}$	• Blast-hole pattern – $B \times S \text{ m}^2$
• Drilling hours / year – H_r	• Sub-grade drilling – $SG \%$
• Drilling rate – $d \text{ min / m}$	• Time to change a drill rod – $e \text{ min}$
• Time to move the drill – $l \text{ min}$	• Time to set the drill – $g \text{ min}$
• Length of feed – $f \text{ m}$	• Overall utilization of drill - E

Estimation of number of drills

Height of the benches is determined considering the shovel capacity intended to be used.

$$\text{Volume of ore to be produced / year} = (T / O_d) \text{ m}^3$$

$$\text{Volume of waste to be handled / year} = ((T \times SR) / W_d) \text{ m}^3$$

$$\text{Total volume of to be handled / year} = ((T / O_d) + (T \times SR) / W_d) \text{ m}^3 = V \text{ m}^3 \text{ (say)}$$

LECTURE NOTES – SURFACE MINING

Volume handled / hole = $B \times S \times BH_t \text{ m}^3$

Total number of holes to be drilled / year = $V / (B \times S \times BH_t) = N_h$ (say)

Depth of blast-hole = $BH_t \times (1 + SG / 100) \text{ m} = BH_d \text{ m}$ (say)

Number of feeds required for a hole = $BH_d / f = N_f$ (say)

Actual number of feeds required for a hole = N_f , N_f is integer
= (Integer part of N_f) + 1, otherwise

Number of feed change required (N_c say) = $N_f - 1$, N_f is integer
= Integer part of N_f , otherwise

Total time to drill one hole = $(l + g + (BH_d \times d) + (N_c \times e)) \text{ min} = t \text{ min}$ (say)

Actual number of holes drilled per hour per drill = $(60 / t) \times E$

Actual number of holes drilled per year per drill = $(60 / t) \times E \times H_r$

Total number of drills required = $N_h / ((60 / t) \times E \times H_r) = N_d$ (say)

Actual number of drills required = N_d , N_d is integer
= (Integer part of N_d) + 1, otherwise

Numerical Example

Required production of iron ore	5 million te / year	5 million te / year
Stripping ratio	0.2 te / te	0.2 te / te
Average in-situ density of ore	3.8 te / m ³	3.8 te / m ³
Average in-situ density of waste	2.8 te / m ³	2.8 te / m ³
Drilling hours per year	300 × 2 × 5 = 3000	300 × 2 × 5 = 3000
Drilling machine	150 mm dia Drill	250 mm dia Drill
Bench height	10 m	12 m
Blast-hole pattern	4 × 5 m ²	5.5 × 7 m ²
Sub-grade drilling	10 %	10 %
Drilling rate	5 min / m	5 min / m
Time to move and set the drill	10 min	10 min
Time to change one drill rod	5 min	5 min
Length of feed	7 m	8 m
Overall utilization of drill	0.6	0.6
Volume of ore to be handled / year	$5 \times 10^6 / 3.8 \text{ m}^3$ = 1315789.5 m ³	$5 \times 10^6 / 3.8 \text{ m}^3$ = 1315789.5 m ³
Volume of waste to be handled / year	$5 \times 0.2 \times 10^6 / 2.8 \text{ m}^3$ = 357142.9 m ³	$5 \times 0.2 \times 10^6 / 2.8 \text{ m}^3$ = 357142.9 m ³

LECTURE NOTES – SURFACE MINING

Total volume to be handled / year	1672932.4 m ³	1672932.4 m ³
Volume handled / hole	$4 \times 5 \times 10 \text{ m}^3$ = 200 m ³	$5.5 \times 7 \times 12 \text{ m}^3$ = 462 m ³
Number of holes required / year	8364.67	3621.07
Depth of hole	$1.1 \times 10 \text{ m} = 11 \text{ m}$	$1.1 \times 12 \text{ m} = 13.2 \text{ m}$
Number of feeds required	$11 / 7 = 1.57 = 2$	$13.2 / 8 = 1.65 = 2$
Number of feed change required	1	1
Total time to drill one hole	$10 + (11 \times 5) + 5 \text{ min}$ = 70 min	$10 + (13.2 \times 5) + 5 \text{ min}$ = 81 min
Actual number of holes drilled per year per drill	$(60 \times 0.6 \times 3000) / 70$ = 1542.85	$(60 \times 0.6 \times 3000) / 81$ = 1333.33
Total number of drills required	$8364.67 / 1542.85$ = 5.422 = 6	$3621.07 / 1333.33$ = 2.716 = 3

LECTURE NOTES – SURFACE MINING

Ripping is considered due to various reasons as stated below:

Increased Productivity: In ripping process there is continuous work going on. This reduces idle time. It also eliminates shifting of machines which is present in blasting operation.

Minimized ground vibration: Drilling and blasting operation includes large ground vibration which affect nearby inhabitation and creates cracks on ground and structures. Ripping and dozing minimizes ground vibration.

Safety: There are chances of generation of fly rocks during blasting. Also chances of misfires are more in blasting operation. Ripping eliminates the chances of generation of fly rocks and misfires, thereby increasing safety of life and properties.

Noise and dust reduction: Drilling and blasting creates a lot of noise and dust which is eliminated by use of rippers.

Product size: Blasting sometimes result in oversize boulders which may require secondary blasting which is a costly affair. But selection of right kind of ripper results in right size of material.

Slope stability: blasting may result in slope failure. But ripping provides better safety and slope stability.

Quality control: In blasting there are chances of dilution of ore. But in ripping operator can easily distinguish between ore and waste. Ripping is helpful in selecting mining.

Cost economics: Ripping doesn't involve involvement of various machineries as in drilling and blasting thereby making the process economical.

Environmental friendly: Ripper provides a pollution free environment to work.

Ripping Techniques

The suitable technique depends on the job conditions. Following factors are to be considered before selecting proper ripping technique.

a) **Ripping speed:** Proper gear and speed plays important role in maximization of ripping production. Generally first gear with a speed of 1-1.5mph gives the economical production with maximum drawbar pull. Excessive speed causes track slippage and rapid undercarriage wear as well as tip wear. Excessive speed heats up the tip thereby decreasing the life of the tip.

b) **Ripping depth:** Ripping depth is a function of job condition, material hardness, lamination thickness, and degree of fracturing. Ripping should be done at maximum depth allowable by the ripper. When considerable stratification is encountered it is preferable to rip at partial depth and remove the material in its natural layer rather than to take out a full ripped layer. Where scrapers are used to lift materials to the dumpers it is advisable to rip to a uniform depth eliminating the hard rocks which can lift off the scraper edged from the ground.

c) **Spacing between the passes:** Optimum spacing between the passes helps in maximizing the production. Closer the spacing, smaller is the chunk size. When full penetration occurs, pass spacing of one-half the tractor width allows the track to move over the material just ripped and increasing the crushing of the same.

d) **Ripping direction:** The ripping direction is decided by the job layout. When scrapers are used to remove material ripping should be done in the direction scraper loading in order to increasing the scraper loading efficiency. When vertical laminations or fractures are found on the rock formations ripping is done across the cuts. Downhill ripping is preferred as it takes the advantage of the tractor weight and horsepower.

Ripping Mechanism

In the process of ripping, the ripper tip is lowered in to the ground by means of hydraulic forces which continues till initial bit penetration is achieved. The initial penetration of the tip or tyne is either occurred through the rock mass, defect planes or combination of both.

When the stress concentration of tip exceeds the compressive strength of the rock it causes shear failure of rock allowing initial tip penetration. As tractor moves the penetration causes tensile failure of rock. In case of jointed rocks the failure takes place due to failure of cohesive force between the structural blocks.